



Using LLM Agents with Llama 3, LangGraph and Milvus

Stephen Batifol | Zilliz

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Speaker



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LLM
& DATA

Easy Setup

Pip-install to start coding in a notebook within seconds.



Reusable Code

Write once, and deploy with one line of code into the production environment



Integration

Plug into OpenAI, Langchain, LlmIndex, and many more



Feature-rich

Dense & sparse embeddings, filtering, reranking and beyond



Milvus is an open-source vector database for **GenAI** projects. `pip install` on your laptop, plug into popular AI dev tools, and push to production with a single line of code.



27K+

GitHub
Stars



2,600

Forks



25M+

Downloads



250+

Contributors

Seamless integration with all popular AI toolkits



VOYAGE AI



RAG

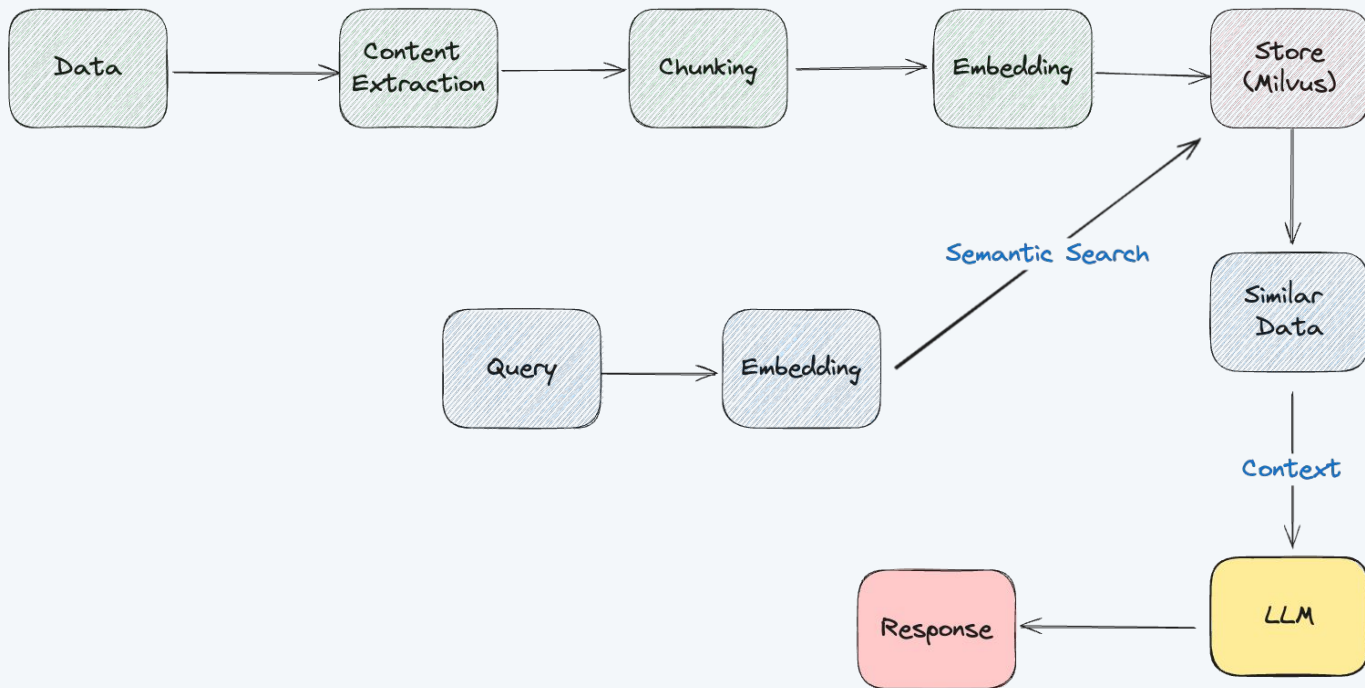
(Retrieval Augmented Generation)

Basic Idea

Use RAG to **force** the **LLM** to **work with your data**
by injecting it via a vector database like **Milvus**

Basic RAG Architecture

RAG Architecture



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Tech Stack



LangChain

- Framework for building LLM Applications
- Focus on retrieving data and integrating with LLMs
- Integrations with most AI popular tools



LangGraph by LangChain

- Build Stateful apps with LLMs and Multi-Agents workflow
- Cycles and Branching
- Human-in-the-Loop
- Persistence

Ollama

- Run LLMs anywhere
- Run Embedding Models



Build Once Deploy Anywhere

| GenAI Application

Milvus Client Code

```
1 client = MilvusClient(...)
2 client.insert(...)
3 client.search(...)
```

| Milvus Lite



| Milvus



02

Agentic RAG

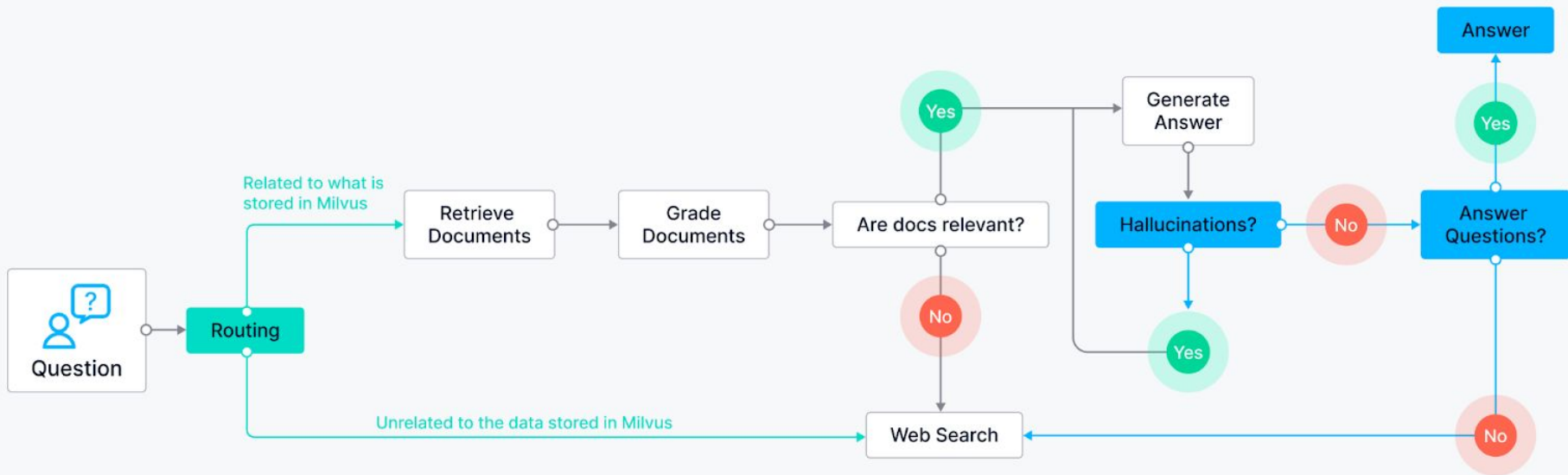
General Ideas

- **Routing: Adaptive RAG**
 - Route Questions to different retrieval approaches
- **Fallback: Corrective RAG**
 - Fallback to web search if docs are not relevant to query
- **Self-Correction: Self-RAG**
 - Try to fix answers with hallucinations or don't address question

General Ideas for Agents

- **Reflection**
 - Self-Correction Mechanism
- **Planning:**
 - The agent doesn't just react to the query
 - Lays out a step-by-step process to retrieve or generate the best answer
- **Tool use**
 - Search for Knowledge Base in Milvus
 - Search the web for more information

General Ideas



Demo!

Questions?



milvus.io



github.com/milvus-io/



@milvusio



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Advanced RAG techniques

Types of RAG Enhancement Techniques

- **Divide & Conquer**
 - **Query Enhancement:** better express or process the query intent.
 - **Indexing Enhancement:** data cleanup, better parser and chunking
 - **Retriever Enhancement:** more retrievers and hybrid search strategy
 - **Generator Enhancement:** prompt engineering and more powerful LLM

Milvus Architecture

